Extreme events: a shift or a change in shape?

Environmental Economics – BSc AIDAMS – Track 2

Abstract

We compare heat and drought distributions in Spain, France, Germany and Sweden using country-level ERA5-Land indices. Fixed historical thresholds, mean-shift benchmarks and quantile regressions distinguish changes in overall levels from additional tail movement. Heat-threshold crossings increase across countries; a shift in the mean accounts for much of the rise. France and Germany also show wider pooled temperature distributions, while evidence of extra upper-tail movement remains imprecise. Drought results vary more across countries and thresholds. The estimates describe historical changes; small tail samples and limitations of the early reanalysis record constrain inference.

1. Question

Have heat and drought extremes become more frequent or more intense in Spain, France, Germany and Sweden, and how much of that change can be described by a shift in the mean? We compare 1961-1990 with 1996-2025 using country-level ERA5-Land data from the Copernicus Interactive Climate Atlas. Our central question is whether the recent distribution resembles the early distribution moved to a new level, or whether its extreme tail has moved further than the mean.

The distinction matters for risk assessment. A fixed historical threshold can be crossed much more often after the mean changes, even if variability around the mean stays constant. More exceedances therefore do not, by themselves, establish that a tail has become heavier. For water managers, agricultural producers and planners, both the level of the climate variable and the distribution around that level matter. Our data measure climate conditions; they do not measure economic losses or identify the benefits of an adaptation policy.

Three references frame the question. IPCC AR6 Chapter 11 documents changes in the frequency and intensity of temperature extremes and assesses their human contribution using evidence beyond simple historical comparisons (Seneviratne et al., 2021, Sections 11.2-11.3). Hansen, Sato and Ruedy (2012) show a shift toward hotter seasonal temperature anomalies and a larger occurrence of exceptionally hot summers. Rhines and Huybers (2013) question the inference of increased variability from those distributions: normalisation, trends and changing station coverage can produce apparent widening. Their result makes the empirical task clear. We must compare tail changes with a shift benchmark, rather than infer a change in shape from an increase in extremes alone. These papers concern seasonal temperatures and different spatial aggregations; our analysis is not a direct replication.

We use two complementary indices: monthly TXx for June-August, measuring the country average of local monthly temperature maxima, and September SPEI-6, summarising April-September climatic water balance. The four countries span southern to northern Europe, allowing us to compare contrasting climates within one dataset. TXx values are expressed relative to each country's early-period mean for the same calendar month. We examine its upper tail at P75, P90, P95 and P99. For SPEI-6, lower values mean drier conditions, so the corresponding thresholds are P25, P10, P5 and P1.

An answer requires three comparisons: the change in the mean and spread; the change in tail percentiles relative to the mean; and the frequency of crossing fixed early-period thresholds. Time trends and quantile regressions provide a further check on whether the centre and tails move at different rates. The results describe these four country-level series over the observed record. They do not attribute individual events to greenhouse gas emissions or predict future extremes.

2. Data

Our data come from the Copernicus Interactive Climate Atlas (https://atlas.climate.copernicus.eu/, version v2.5), produced by the Copernicus Climate Change Service (C3S). The Atlas is built on the gridded dataset that C3S publishes on the Climate Data Store (C3S, 2024b). We downloaded the series through the Atlas's regional-information export on 29 October 2026. All series come from ERA5-Land, a land-surface reanalysis that ECMWF produces for C3S by re-running the land component of the ERA5 model on a 0.1° regular grid; the Atlas retains this native resolution (C3S, 2023). We study four European countries along a south-north gradient: Spain, France, Germany and Sweden. Every series is annual and covers 1950 to 2025, giving 76 observations per series. A validation script (scripts/check_raw.py) confirms that all 16 files share the same Atlas version and dataset, that each year from 1950 to 2025 appears exactly once, and that no observation is missing. We therefore carry out no imputation or interpolation. The label "1961-1990" in the file headers refers to the period the Atlas highlights in its charts; it does not restrict the data.

We use two indices. The first, TXx, is the monthly maximum of daily maximum near-surface air temperature, in degrees Celsius. In each grid cell, the daily maximum is the highest of the hourly 2 m temperatures of that day, and the Atlas takes the highest of these daily maxima in a given month (C3S, 2023). It then averages these cell values over the country's territory, weighting each cell of the regular latitude-longitude grid by the cosine of its latitude; since ERA5-Land covers land only, sea cells do not enter the average (C3S, 2024a). Our country-level TXx is therefore the average across the country of each location's hottest day of the month, not the single hottest value recorded anywhere in the country. We use TXx for June, July and August. Pooling the three summer months gives 90 observations per 30-year period instead of the 30 that a single month would provide. This matters because our question concerns the tails of the distribution, which are poorly estimated in small samples. The cost is that the three months are not independent draws from one distribution, as August is typically hotter than June. Any pooled comparison must allow for this seasonal difference.

The second index is the Standardised Precipitation-Evapotranspiration Index over a six-month window (SPEI-6). For each grid cell, the balance between precipitation and potential evapotranspiration (PET) is accumulated over the preceding six months. It is then transformed, by fitting a probability distribution, into standard deviations from the mean of a reference period of 1971-2005. The Atlas documentation is inconsistent on this point: the dataset documentation gives 1971-2010 (C3S, 2023), while the v2.5 changelog gives 1971-2005 in the context of projections (C3S, 2024a). The Atlas information panel for SPEI-6, however, states explicitly that the reference period is 1971-2005 "for observations, reanalysis, and projections" (verbatim copy in data/atlas_info_spei6.txt), and we report this value. Potential evapotranspiration is computed with the Hargreaves formulation (Hargreaves and Samani, 1985), which the Atlas has used since version 2.0 (C3S, 2024a), implemented with the xclim library (C3S, 2023). The index has no units; negative values indicate conditions drier than the reference climate and positive values wetter ones. As with TXx, the Atlas averages the grid-cell values over each country. We keep only the September value in each year. Because the window is six months long, the September value summarises the water balance from April to September, that is, the whole growing and warm season. Taking one month per year also avoids the strong autocorrelation that consecutive monthly SPEI-6 values would introduce, as they share five of their six months. Each year then contributes one non-overlapping observation of warm-season dryness.

Data limitations

These data have several limitations. First, ERA5-Land is a reanalysis, not a set of station observations. It combines a numerical model with assimilated observations, and its representation of extremes depends on the model's physics as well as on the observations. A specific concern affects the start of our sample: the Atlas disables ERA5-Land trends before 1978 because of a potential inconsistency in the ERA5 forcing used for that period (C3S, 2024a). Our early period overlaps these years, so part of any difference between the early and recent distributions could be a data artefact rather than a change in climate.

Second, averaging over a country smooths local extremes. A heatwave or drought that affects only part of the territory is diluted in the national mean, so our series understate the severity of the most extreme local events, and the smoothing is likely to be stronger for large or climatically diverse countries such as France and Spain. In addition, the Atlas documentation does not state how grid cells that straddle a national border are treated (C3S, 2024a), so the exact spatial footprint of each country average is not known.

Third, SPEI depends on modelling choices: the reference period sets the zero of the scale, and the PET formulation determines how strongly rising temperatures translate into apparent drying. A different reference period or PET method could shift the series or change their trends.

Fourth, standardisation is done per grid cell, and averaging standardised values across cells reduces their variance. The country-level SPEI-6 series have standard deviations between 0.754 and 0.902, all below the unit variance of a cell-level index, and their 1950-2025 means are negative, between -0.285 and -0.037. As a result, the Atlas's fixed threshold of -1.5 for ”severe” drought is rarely crossed at the country level, and a count of exceedances of that threshold would carry little information. We therefore define extremes with percentile thresholds taken from each country's own distribution.

Finally, the tails contain few observations by construction. With 76 years per series, or 90 observations per 30-year period for pooled TXx, the 90th or 95th percentile within a period is estimated from only a handful of values. Estimates of tail shape are thus imprecise. Section 5 reports bootstrap intervals that keep the three months of a summer together when resampling; these do not account for dependence across years.

3. Summary statistics

We compare 1961-1990 (early, E) with 1996-2025 (recent, R). TXx pools June-August anomalies relative to the early mean of the same country and month. SPEI-6 is used as published, one September value per year. The distributions contain 90 TXx values or 30 SPEI-6 values per country and period. Percentiles use linear interpolation.

Table 1. Summary statistics: early and recent distributions.

Index	Country	Period	N	Mean	SD	Min	P5	P95	Max
TXx	ESP	E	90	0.00	1.61	-5.05	-2.86	2.03	4.06
TXx	ESP	R	90	2.10	1.74	-2.16	-0.69	5.31	5.89
TXx	FRA	E	90	0.00	1.83	-3.57	-2.73	3.16	4.27
TXx	FRA	R	90	2.44	2.35	-4.04	-1.02	6.35	8.21
TXx	GER	E	90	0.00	1.78	-4.63	-2.95	2.89	4.01
TXx	GER	R	90	1.85	2.37	-4.12	-1.91	5.29	6.74
TXx	SWE	E	90	0.00	2.06	-4.85	-3.76	3.18	4.61
TXx	SWE	R	90	0.63	2.15	-4.45	-2.76	4.18	4.98
SPEI-6	ESP	E	30	0.15	0.75	-1.02	-0.89	1.44	2.30
SPEI-6	ESP	R	30	-0.62	0.70	-1.82	-1.63	0.59	0.80
SPEI-6	FRA	E	30	0.05	0.86	-1.70	-1.40	1.15	1.25
SPEI-6	FRA	R	30	-0.56	0.77	-1.98	-1.56	0.41	0.85
SPEI-6	GER	E	30	0.21	0.82	-1.90	-1.04	1.31	1.70
SPEI-6	GER	R	30	-0.36	0.91	-2.80	-1.95	0.82	0.97
SPEI-6	SWE	E	30	-0.10	0.68	-1.99	-1.27	0.99	1.26
SPEI-6	SWE	R	30	0.07	0.80	-1.89	-1.15	1.42	1.56

TXx: anomalies in °C. SPEI-6: dimensionless, negative = drier. E: 1961-1990; R: 1996-2025. Country codes: Spain (ESP), France (FRA), Germany (GER), Sweden (SWE). Source: C3S Atlas v2.5, ERA5-Land, accessed 29 October 2026; our calculations. Full quantiles: outputs/summary_stats.csv.

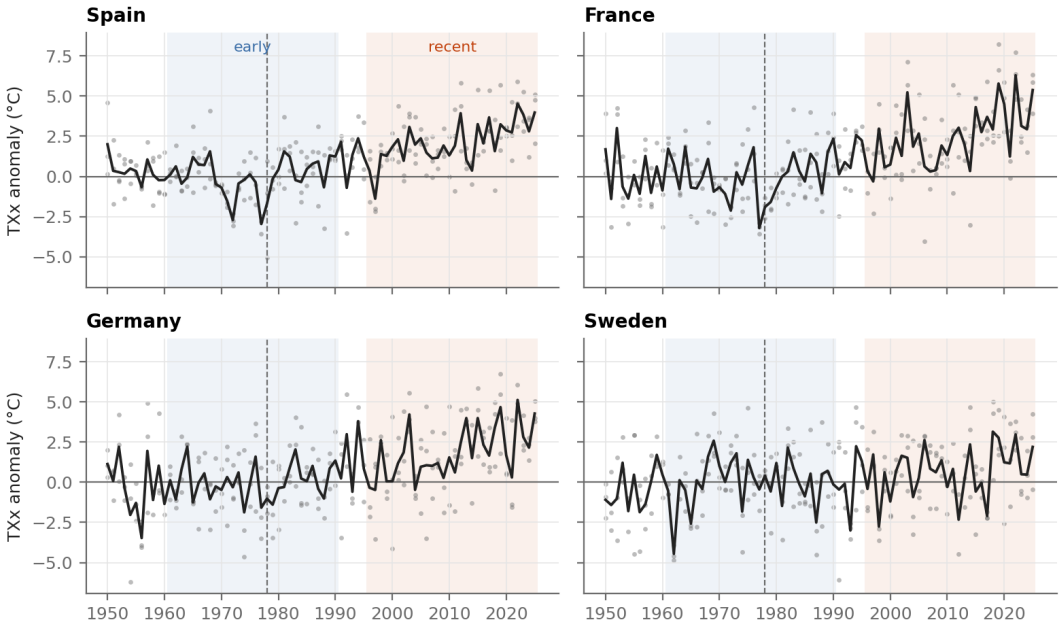
Table 2. TXx reference levels: 1961-1990 mean, °C.

Country	June	July	August
ESP	30.14	33.31	32.90
FRA	26.85	29.59	28.97
GER	26.86	28.29	28.19
SWE	23.27	24.06	22.27

TXx is higher in the recent period in all four countries. Mean shifts are larger in Spain, France and Germany than in Sweden. The P95 rises more than the mean in each country; Section 5 assesses the precision of that extra movement. France and Germany also show wider pooled distributions. SPEI-6 shifts toward drier conditions in Spain, France and Germany, while Sweden changes little. Tail precision is limited: each period has only five TXx observations above its own P95 and two SPEI-6 observations below its own P5.

4. Figures

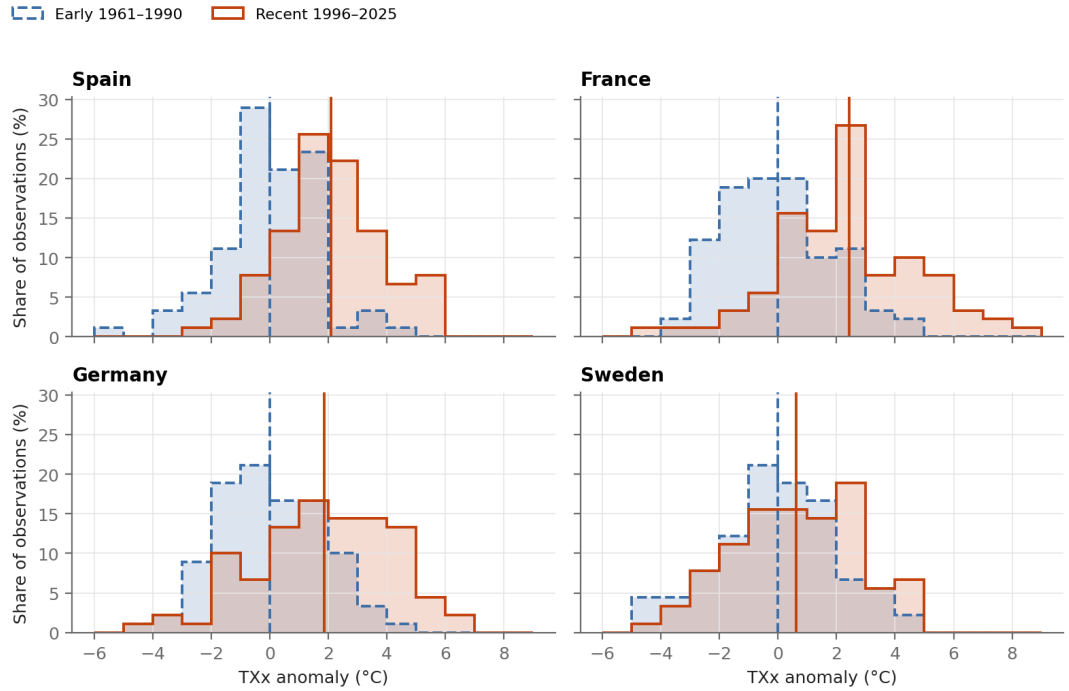
Figure 1. Summer maximum temperature (TXx) anomaly, 1950–2025



Dots: June, July and August values; line: their yearly mean. Anomaly in °C from the 1961–1990 mean of the same month. Shaded: the two compared periods. Dashed line: 1978, before which ERA5-Land may be less consistent. Source: Copernicus Interactive Climate Atlas v2.5, ERA5-Land, accessed 29 Oct 2026.

Figure 1. Country-average summer TXx anomalies show upward trends with substantial year-to-year variation.

Figure 2. Distribution of summer TXx anomalies, early vs recent period



Share of the 90 June–August values per period falling in each 1 °C bin. Vertical lines: period means. Anomaly from the 1961–1990 mean of the same month. Source: Copernicus Interactive Climate Atlas v2.5, ERA5-Land, accessed 29 Oct 2026.

Figure 2. Recent TXx distributions lie further toward hot anomalies in Spain, France and Germany; Sweden's distributions overlap more.

4. Figures

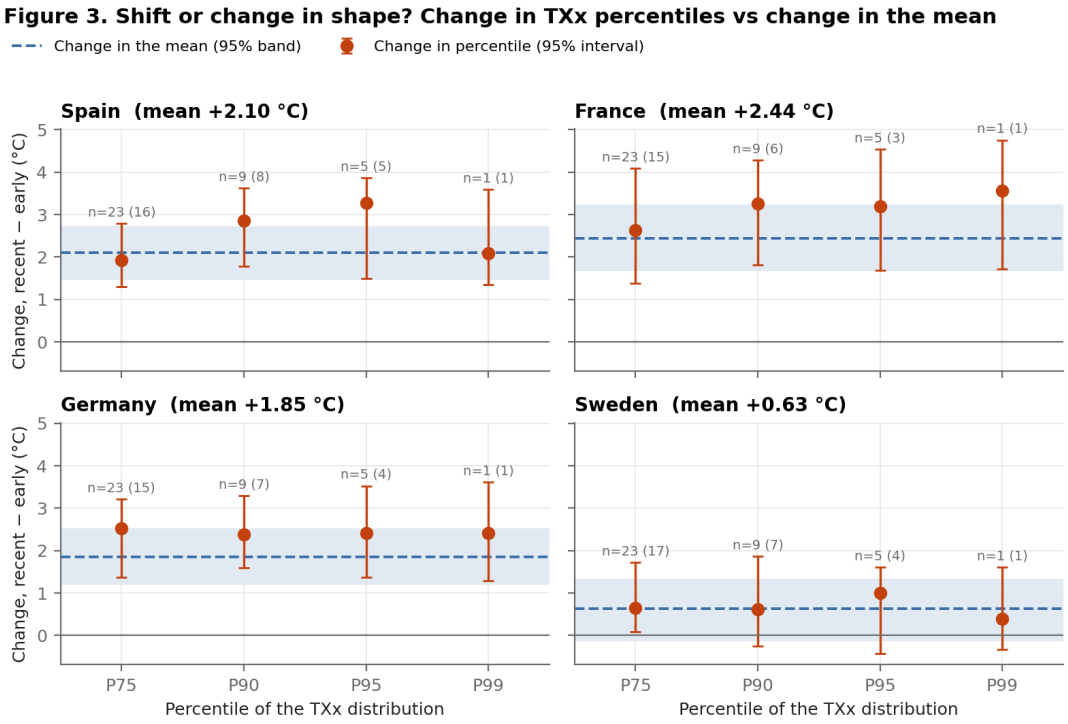


Figure 3. TXx upper percentiles rise, but their extra movement relative to the mean is imprecise in every country.

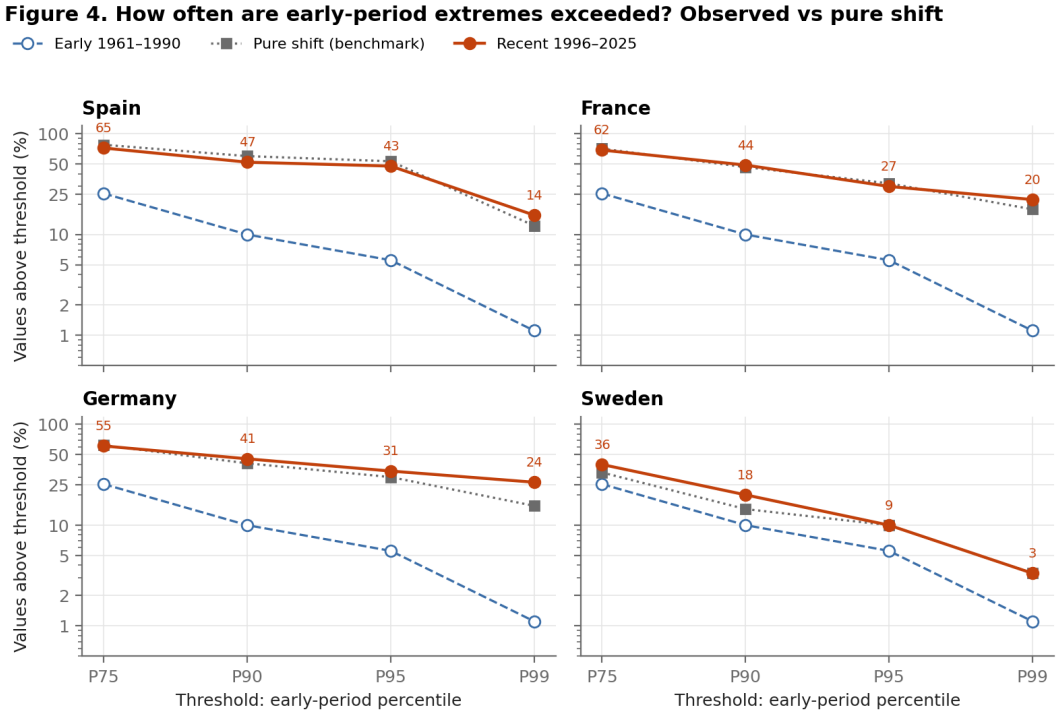
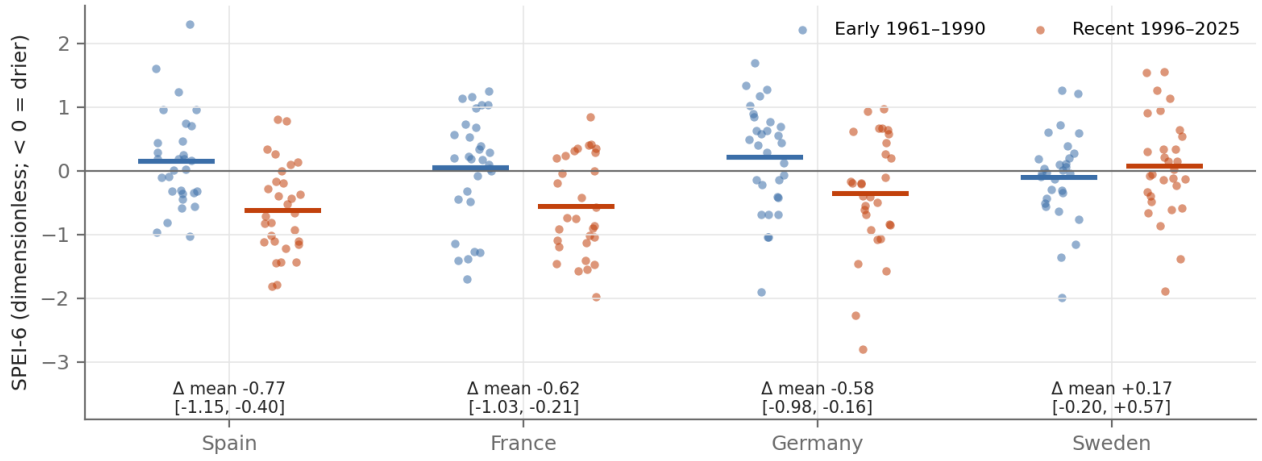


Figure 4. Heat exceedances increase at every threshold; the shift benchmark accounts for much of the rise, with a larger gap at Germany's P99.

4. Figures

Figure 5. Drought: September SPEI-6, early vs recent period



Each dot is one year's September SPEI-6 (April–September water balance), 30 per period. Bars: period means. Δ mean: recent − early with 95% bootstrap interval. Reference climate of the index: 1971–2005. Source: Copernicus Interactive Climate Atlas v2.5, ERA5-Land, accessed 29 Oct 2026.

Figure 5. Mean September SPEI-6 falls in Spain, France and Germany; Sweden shows no clear change between periods.

Each point represents one September value, covering April-September climatic water balance; there are 30 observations per country and period. Mean changes are -0.77 in Spain, -0.62 in France and -0.58 in Germany, with 95% intervals excluding zero, and +0.17 in Sweden, with an interval including zero. Only two observations lie below each period's own P5. The estimates therefore provide limited information about changes in the most severe drought tail.

The fixed-threshold comparison and the quantile regressions in Section 5 assess whether these changes extend beyond a shift in the mean. A national index can change substantially without identifying a change in local event duration or an economic loss.

5. Econometrics

5.1. Fixed thresholds and a shift benchmark

We set each country's thresholds on 1961-1990 and apply the same values to 1996-2025. Frequency is the share of observations above the TXx threshold or below the SPEI-6 threshold, not a count of distinct heatwaves or drought episodes. Let the mean shift be

$$\Delta = \bar{y}_R - \bar{y}_E.$$

The shift benchmark adds delta to every early observation and recounts exceedances. It preserves the early distribution's shape and uses the observed mean change; it is a descriptive benchmark, not an independent forecast.

Table 3. Threshold sensitivity: early count / recent count / shift-benchmark count.

Index	Country	P75 / P25	P90 / P10	P95 / P5	P99 / P1
TXx	Spain	23 / 65 / 70	9 / 47 / 54	5 / 43 / 48	1 / 14 / 11
TXx	France	23 / 62 / 64	9 / 44 / 42	5 / 27 / 29	1 / 20 / 16
TXx	Germany	23 / 55 / 56	9 / 41 / 37	5 / 31 / 27	1 / 24 / 14
TXx	Sweden	23 / 36 / 30	9 / 18 / 13	5 / 9 / 9	1 / 3 / 3
SPEI-6	Spain	8 / 20 / 21	3 / 16 / 15	2 / 12 / 11	1 / 11 / 11
SPEI-6	France	8 / 18 / 16	3 / 6 / 6	2 / 6 / 6	1 / 1 / 6
SPEI-6	Germany	8 / 15 / 13	3 / 9 / 9	2 / 6 / 6	1 / 2 / 1
SPEI-6	Sweden	8 / 7 / 5	3 / 3 / 3	2 / 2 / 1	1 / 1 / 1

Each TXx count is out of 90 country-month observations per period; each SPEI-6 count is out of 30 September observations. Thresholds are early-period percentiles, with strict inequalities and linear interpolation. Source: C3S, Copernicus Interactive Climate Atlas v2.5, ERA5-Land, accessed 29 October 2026; outputs/exceedance.csv.

Heat exceedances increase at every threshold in every country. At P95, Spain rises from 5 to 43 observations, France from 5 to 27, Germany from 5 to 31 and Sweden from 5 to 9. These recent exceedances come from 23, 19, 20 and 7 distinct summers, respectively. The early P95 rate is 5/90, or 5.6%, rather than exactly 5%. The shift benchmark gives 48, 29, 27 and 9 exceedances. Thus much of the increase can occur without a change in shape. Germany's P99 is the clearest discrepancy: 24 observed against 14 under the benchmark. No confidence interval is estimated for these count gaps, so they are not formal tests of a pure shift.

Drought results depend more on the threshold. Spain has more dry observations at all four cut-offs. France and Germany have more at P25, P10 and P5, but France's P1 count stays at one. Sweden's counts change little. At the headline P5, recent counts are 12, 6, 6 and 2; the shift benchmark gives 11, 6, 6 and 1. P99 for TXx and P1 for SPEI-6 each have only one early observation beyond the line, making frequency ratios particularly unstable.

5.2. Tail intensity relative to the mean

We measure departure from a shift by

$$S_p = (Q_{R,p} - Q_{E,p}) - \Delta,$$

where Q is the percentile at p and E and R denote the early and recent periods. A positive S_p means extra movement toward heat for TXx; a negative value means extra movement toward dryness for SPEI-6. For TXx at P95, S_p is 1.18 °C in Spain, 0.75 °C in France, 0.55 °C in Germany and 0.37 °C in Sweden. Their 95% bootstrap intervals are [-0.40, 1.72], [-0.54, 1.70], [-0.31, 1.47] and [-0.84, 1.03] °C. All include zero, as do the corresponding intervals at the other thresholds. This is limited evidence about tail departures, not proof of identical shapes. France and Germany nevertheless show wider pooled distributions: their SD changes are 0.52 °C [0.07, 0.93] and 0.59 °C [0.18, 0.98].

Intervals use 2,000 bootstrap draws, resampling years independently within each period and keeping a summer's three months together. At each period's own P95, only five TXx values lie above the line; at its own P5, only two SPEI-6 values lie below it. We do not detrend within periods, so changes in pooled spread can include changes in the mean within a period (Rhines and Huybers, 2013). Resampling single years preserves dependence within summers but does not preserve dependence across years; intervals are pointwise and do not adjust for multiple comparisons.

5.3. Mean and quantile trends

For each country and index, OLS estimates

$$y_t = \alpha + \beta \frac{t - 1950}{10} + u_t.$$

Here t is the calendar year; y_t is the annual June-August mean TXx anomaly (°C) or September SPEI-6 (dimensionless); alpha is the fitted level in 1950 and u_t the residual. Beta measures change per decade. Newey-West HAC standard errors use three annual lags. Quantile regression estimates

$$Q_\tau(y \mid t) = \alpha_\tau + \beta_\tau \frac{t - 1950}{10}$$

on pooled monthly TXx anomalies or annual SPEI-6. We fit tau = 0.05, 0.10, 0.25, 0.50, 0.75, 0.90 and 0.95; Table 4 reports the median and relevant extreme tail.

Table 4. Trends per decade; standard errors in parentheses.

Index	Country	OLS 1950-2025	OLS 1978-2025	QR median	QR tail
TXx	Spain	0.440 (0.081)	0.722 (0.085)	0.452 (0.073)	0.511 (0.114)
TXx	France	0.550 (0.096)	0.947 (0.117)	0.562 (0.088)	0.558 (0.173)
TXx	Germany	0.459 (0.078)	0.719 (0.103)	0.466 (0.086)	0.428 (0.158)
TXx	Sweden	0.215 (0.068)	0.296 (0.118)	0.245 (0.114)	0.226 (0.073)
SPEI-6	Spain	-0.103 (0.048)	-0.243 (0.037)	-0.131 (0.052)	0.017 (0.070)
SPEI-6	France	-0.067 (0.053)	-0.248 (0.067)	-0.202 (0.095)	0.024 (0.036)
SPEI-6	Germany	-0.079 (0.053)	-0.223 (0.079)	-0.069 (0.072)	-0.126 (0.135)
SPEI-6	Sweden	0.027 (0.036)	-0.015 (0.064)	-0.015 (0.034)	0.112 (0.132)

OLS: 76 annual values for 1950-2025 and 48 for 1978-2025, with HAC errors. Quantile regression: 1950-2025, 228 TXx observations or 76 SPEI-6 observations; 499 whole-year bootstrap draws for standard errors. Tail means tau = 0.95 for TXx and 0.05 for SPEI-6. Units: °C/decade for TXx, index units/decade for SPEI-6. Source: as Table 3; outputs/ols_trends.csv and outputs/quantile_trends.csv.

Full-record mean TXx trends are positive in all four countries. France's 0.55 °C per decade means a fitted rise of about half a degree per decade in the average of the three monthly maxima, not in annual mean temperature. Its median and P95 slopes are both about 0.56 °C per decade. In every country, the P95-minus-median slope interval includes zero; all SPEI-6 P5-minus-median intervals also include zero. These direct contrasts provide no clear evidence that the relevant extreme tail trends faster than the centre. They have low precision: only 12 observations lie above each fitted TXx P95 line and five below each SPEI-6 P5 line. Some quantile fits reach the iteration limit; series totals range from 1 to 10 across 3,500 point and bootstrap fits, so affected estimates remain approximate.

Re-estimating OLS from 1978 keeps positive mean TXx trends in every country and gives negative SPEI-6 trends in Spain, France and Germany whose 95% intervals exclude zero. Sweden's SPEI-6 interval includes zero. This addresses the documented concern about early ERA5-Land forcing only for mean trends: neither quantile regressions nor the early-recent comparison are re-estimated on a wholly post-1978 sample. Different slopes across these nested samples are not a formal test of acceleration. All regressions describe association with time; they do not identify a causal effect.

6. What we can and cannot conclude

The clearest result is a shift toward hotter summer maxima in Spain, France and Germany. Between 1961-1990 and 1996-2025, mean TXx anomalies rise by 2.10 °C, 2.44 °C and 1.85 °C, respectively. Sweden's estimated rise is smaller, 0.63 °C, and its period-difference interval includes zero, although its full-record mean trend is positive with a HAC interval excluding zero. A comparison of two periods and a trend over the full record answer different questions and need not have the same precision.

Heat observations cross fixed historical thresholds more often in all four countries, whichever of P75, P90, P95 or P99 defines the extreme. A shift of the early distribution by the observed mean change reproduces much of that increase. The upper percentiles also rise, but their extra movement relative to the mean is imprecise. The quantile regressions similarly do not establish a faster P95 trend than the median. The evidence therefore supports a substantial shift, while leaving moderate changes in the upper tail unresolved. France and Germany have wider pooled distributions, so the results do not justify a blanket claim that shape is unchanged.

For drought, mean September SPEI-6 falls in Spain, France and Germany, indicating a drier April-September climatic water balance. Sweden shows no clear mean drying. Spain's increase in dry-threshold crossings persists across all four cut-offs. The conclusion is less uniform elsewhere: France has more crossings at P25, P10 and P5, but no increase at P1. We find no clear extra deterioration of the P5 relative to the mean. With only 30 observations per comparison period, this is weak evidence about the most severe droughts.

Our method combines distribution comparisons, fixed-threshold counts, a shift benchmark, OLS trends and quantile regression. The results are descriptive changes and associations with time. A time coefficient absorbs many processes; it cannot isolate greenhouse gas forcing, land-use change or a particular economic mechanism. Causal attribution would require a credible counterfactual, such as climate-model ensembles with and without human forcing, together with evidence on model performance and observational uncertainty. The broader attribution evidence assessed by Seneviratne et al. (2021) does not turn our regressions into an attribution design.

Three limits are central. National averages smooth local extremes, and these indices do not count heatwave duration or observed crop and income losses. The most extreme percentiles rest on very few observations, while our bootstrap does not preserve dependence across years. Finally, the early baseline overlaps the documented pre-1978 ERA5-Land concern; the post-1978 mean-trend check cannot validate all early-period tail results. The practical implication is still clear: a change in the mean can substantially raise exposure to historically unusual conditions. The data support that conclusion more firmly than a claim that the extreme tails have systematically become heavier.

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Statement on the use of AI

We used Claude Code and OpenAI Codex for code execution and help with concrete system design. Codex also assisted with formatting of the paper. Numerical statements were checked against the repository's computed outputs. The authors remain responsible for reviewing the submitted text, verifying its references and explaining the methods and results.